MIL-T-83928A (USAF)
9 JULY 1980
SUPERSEDING
MIL-T-83928 (USAF)
31 MAY 1967

MILITARY SPECIFICATION
Test Set, Guided Missile Launchers, SM-419/A

1. SCOPE

1.1 This specification covers the SM-419/A Test Set hereinafter referred to as "Test Set." (For operational use refer to Section 6 of this specification).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS:

Federal

PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Walled

Military

MIL-P-116	Preservation, Methods of
MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Non-Rigid
MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible)
MIL-C-5015	Connectors, Electrical "AN" Type
MIL-L-10547	Liners, Case and Sheet, Water Vapor Proof or Waterproof
MIL-M-13231	Marking or Electronic Items
MIL-W-16878	Wire, Electrical (Insulated, High Temperature)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specifications for

Standards

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-756	Reliability Prediction
MIL-sTD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Require-ments for

DRAWINGS

Air Force

AF64D22408	Cable Assemblies, Special Purposes, Electrical
AF7032753	Nameplate, Test Equipment, SM-419/A, AIM-9 Missile
AF7032757	Plate, Mount, Test Equipment, SM-419/A, AIM-9 Missile
AF7032758	Mount Plate Assy., Test Equipment, SM-419/A, AIM-9 Missile
AF7032760	Schematic, Test Equipment, SM-419/A, AIM-9 Missile
AF7130520	Power Cable Assy., Test Equipment, SM-419/A, AIM-9 Missile
AF7130534	Simulator, Arming Circuit, Aircraft, SM-419/A, AIM-9 Missile, Case Assembly

PUBLICATIONS

Military

ANA Bulletin 438	Age Controls for Age Sensitive Elastomeric Items
MIL-HDBK-217	Reliability, Stress and Failure Rate Data for Electronic Equipment

(Copies of applicable specifications, drawings and publications required by contractors in connection with specific procurement functions may be obtained

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing of three Test Sets.

3.2 Parts and Material. In the selection of parts and material, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Parts and material as approved by MIL-T-28800 shall be given first consideration.

b. When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements, substitute material and parts shall be used only with the approval of the cognizant AIM-9 Missile Service Engineering Activity in accordance with MIL-T-28800.

3.2.1 All electrical connectors shall be in accordance with MIL-C-5015. All cable and wire shall conform to MIL-C-3432, Class MG, Flexibility F. Cable sleeving shall be used and shall meet the requirements of MIL-T-631, Type F, Subform Ua, Grade A, Class I, Category I. Black sleeving should be used to provide a neat appearance.

3.3 Design and Construction.

3.3.1 Configuration. The dimensions and physical layout of the Test Set shall be in accordance with Air Force (AF) Drawings 7032757 and 7032758.

3.3.2 Design References. Drawings 7032757, 7032758, 7032760, 7130520 and 7130534 shall also be used as references and guides for design of the Test Set. This equipment shall meet all applicable design and construction requirements of MIL-T-28800 for Class 2 equipment, except as otherwise specified herein.

3.3.3 Assemblies. The following assemblies are covered by this specification:

3.3.3.1 Test Set. The Test Set shall include the following assemblies:

a. Aircraft Arming Circuit Simulator with Case, SM-419/A.

b. Power Cable Assy., Test Equipment, SM-419/A.

3.3.3.2 Special Purpose Assemblies. Optional special purpose assemblies are:

a. Special Purpose Cable Assembly CX11066/U for Launcher Adapter Type VI.

b. Special Purpose Cable Assembly CX11068/U for Testing Red Dog Launcher.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

c. Special Purpose Cable Assembly CX11069/U for Testing Power Supplies used in Red Dog Launchers.

d. Special Purpose Cable Assembly CX11070/U for Testing AERO-3B Launcher and all Power Supplies used in AERO-3B and Type IX and Type IXA Pylons.

e. Special Purpose Cable Assembly CX11071/U for Testing Type IX and Type IXA Pylons.

f. Special Purpose Cable Assembly CX12938/U for Testing F-5 Launcher power Supply, Type PP4506/A and PP4507/A.

g. Special Purpose Cable Assembly CX12937/U for Testing F-5 Launchers, Types LAU-57/A and LAU-58/A.

h. Special Purpose Cable Assembly CX12939/U for Testing LAU-7A/A-1.

3.3.4 Electrical Requirements. The equipment covered by this specification shall facilitate the testing of all AIM-9 launcher power supplies by simulating simultaneously the armament selector, switching and power application of fire control circuits of each aircraft that utilizes the AIM-9 Missile. Test Set must also be capable of operating on any input voltage combinations of 24 to 32 Volts Direct Current (VDC) and 102 to 120 Volts Root Mean Square (VRMS) at 320 to 522 cycles Alternating Current (AC). Chassis and control panel of the SM-419/A Simulator shall be wired Per Drawing 7032760 in accordance with MIL-W-16878 while the special purpose cables shall be wired per Drawing 7032758.

3.3.5 Reliability.

3.3.5.1 The equipment shall have Mean-Time-Between-Failure (MTBF) of 500 hours at 90% confidence. Failure is defined as the inability of the Test Set to meet all of the specified functional or mechanical requirements.

3.3.5.2 Service Life. The Test Set shall be capable of operating for a total of 7500 complete cycles with minor repairs/replacement of parts.

3.3.6 Case. The case shall be a combination case as specified in MIL-T-21200.

3.3.7 Nameplate. Nameplate shall conform to MIL-N-18307. Part identifications shall conform to Drawings 7032758 and AF64D22408 and MIL-M-13231.

3.4 Service Environments. The Test Set shall operate without degradation of performance under any combination of service conditions specified under Table I of MIL-STD-810 for general base sheltered equipment.

3.5 Workmanship. The Test Set including all parts and accessories therein,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

shall be finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, thoroughness of soldering, wiring, marking of parts and assemblies and freedom of parts from burrs and sharp edges. All Test Sets procured under this specification shall be subject to Government rejection for poor workmanship.

3.6 Data Submittal. Necessary drawings, reports and data to be furnished by the contractor will be specified in the contract.

3.7 Safety. A safety program shall be established and implemented in accordance with the requirements of MIL-STD-882.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in contract or this specification, the contractor is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections or tests are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Operation Reliability. The contractor may submit an estimate to the procuring activity of the reliable operating and service life of the Test Set based on his analysis of the component parts, in accordance with MIL-STD-756 in lieu of actual testing in accordance with MIL-STD-781 to demonstrate the requirements of 3.3.5. Source of component reliability/life data may be either the supplier or the contractor or MIL-HDBK-217 and computations thereof shall be verified per Government documents.

4.3 Preproduction Tests. Preproduction tests are intended to provide assurance that the design and operation of the Test Set will meet the requirements. Successful completion of the test program does not relieve the contractor of the responsibility to furnish equipment that meets the performance requirements of this specification.

4.3.1 Preproduction Equipment. Preproduction tests shall be accomplished on three items representative of the production Test Sets. The three items shall be dientified with nameplates and compel identification data in the same manner as production test sets, including serial number identification.

4.3.2 Test Equipment Personnel. The contractor shall furnish all equipment and shall be responsible for accomplishing all of the preproduction tests. All inspection and testing shall be under the supervision of a Government representative.

4.3.3 Preproduction tests shall consist of the following:

a. Examination of Product. The Test Set shall be examined to determine conformance to this specification with respect to materials, workmanship and

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

general requirements.

b. Insulation Resistance Test. A high electrical potential of 500 VDC shall be applied between each pair of connector pins open circuited to each other. The same potential shall also be applied from ground to each conenctor pin open circuited to ground. Pilot light bulbs shall be removed wherever a pair of pins may be open circuited from each other thereby.

c. Continuity Tests. A continuity test shall be conducted to demonstrate compliance with the continuity requirements of Drawings 7032760 and AF64D22408.

d. Operational Tests. The Test Set shall be given an operational test to verify proper operation at all intended uses given in Section 6.

e. Environmental Tests. The equipment shall be environmentally testes in accordance with those tests listed in Table 1, Note 2A of MIL-STD-810.

4.4 Service Approval Tests. Service approval tests shall be conducted by the procuring activity on the preproduction units. No additional equipment shall be accepted until approval of the preproduction samples. Production of additional equipment prior to samples approval shall be at the contractor's own risk.

4.5 Production Acceptance Tests.

4.5.1 Production Reliability Tests. Production reliability tests shall be conducted on each Test Set in accordance with 4.3.3a, 4.3.3b, 4.3.3c and 4.3.3d.

4.5.2 Environmental Sampling Tests. During the course of production, Test Sets shall be subjected to the vibration tests specified in MIL-STD-810, Method 514, Equipment Class 7.

a. Sampling Procedures. Sampling procedures shall be provided per MIL-STD-105 for special inspection Level S-3, single sampling, normal inspection. Acceptance Quality Level (AQL) value of 4.0 for major defects. An operation test (4.3.3d) and an insulation test (4.3.3b) shall be performed before the start of and after completion of the vibration test. Quantitive data shall be recorded.

b. Inspection Lot. An inspection lot shall consist of maximum of one month's production but not more than 75 Test Sets.

c. Unit of Product and Sample Selection. The unit of product for sampling purposes shall be one complete Test Set. Samples shall be selected in a random manner from the production lot. The lot identity shall be maintained during performance of this test and no units from the lot shall be delivered prior to determination of acceptability of the sample.

d. Defect Classification. Each unit which fails to meet one or more of

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

the specified functional or mechanical requirements as a result of the vibration test shall be deemed to be one major defect. No defects resulting from this test shall be classified as critical or minor. Each such defect shall be repaired and the test stared over, except where the vibration period need not be repeated when it is certain that the failure was not a result of vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. Environmental testing shall be done in periods of not less than three hours each.

4.6 Inspection of Preparation for Delivery. Inspection shall be made to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

a. Level A. Each Test Set shall be packaged in accordance with Method IIb of MIL-P-116.

b. Level C. Test Sets shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

a. Level A. Test Sets packaged in accordance with 5.1.a shall be packed in containers conforming to PPP-B-591 Fiberboard, Wood-Cleated, Overseas Type, PPP-B-601 Wood, Cleated-Plywood, Overseas Type, PPP-B-621 Wood, Nailed and Lock-Corner, Class 2, PPP-B-636 Fiberboard, Weather Resistant or PPP-B-640 Fiberboard, Corrugated Triple-Wall, Class 2. Boxes shall be sealed, closed and strapped for overseas shipment in accordance with applicable box specification and appendix thereto. Wood or cleated boxes shall be lined with a sealed waterproof case defects. An operation test (4.3.3d) and an insulation test (4.3.3b) shall be performed before the start of and after completion of the vibration test. Quantitive data shall be recorded.

b. Level B. Test Sets packaged in accordance with 5.1.a shall be packed in containers conforming to PP-B-591 Fiberboard, Wood-Cleated, Domestic Type, PPP-B-601 Wood Cleated-Plywood, Domestic Type, PPP-B-621, Wood, Nailed and Lock-Corner, Class 1, PPP-B-636 Fiberboard, Domestic or PPP-B-640 Fiberboard, Corrugated, Triple-Wall, Class 1. Boxes shall be sealed and closed in accordance with the requirements of the applicable box specification. The gross weight of wood, wood-cleated or triple-wall corrugated boxes shall not exceed 250 pounds. Fiberboard boxes shall not exceed weight limitation of the applicable box specification.

c. Level C. Test Sets shall be packed in containers in a manner to insure acceptance and safe delivery at destination. Containers shall comply with the consolidated freigt classification rules or other regulations as applicable to the mode of transportation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to any special marking required by the contractor or order, marking for shipment shall be in accordance with MIL-STD-129.

5.4 Cure Dates. The container shall comply with the requirements of ANA Bulletin 438 as applicable.

6. NOTES

6.1 Intended Use. The equipment covered by this specification is intended to facilitate the test of all AIM-9 launcher power supplies by simulating simultaneously the armament selector, switching and power application of aircraft fire control circuits of any aircraft utilizing the AIM-9 Missile.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, numbers and date of this specification.

b. Levels of packaging and packing required (See 5.1 and 5.2).

c. Serial number applicable to contract quantities of Test Sets.

CUSTODIAN:
Air Force - 99

PREPARING ACTIVITY
Air Force - 99

AIR FORCE PROJECT NO
4935-F019

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
3. DOCUMENT TITLE		

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 20200328T09:58Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.